

Designing a Role-play Game for Learning Taiwan History and Geography

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Abstract—In recent years, digital games have been widely used in all fields for learning. Learners are immersed in the context and content, and have increased willingness to participate learning activities which implies the enhancement of learning motivations. The game built in this research was done with Oblivion game engine. First, we design a role-play game script according to the anti-Japanese war history and geography of Southern Taiwan. And then we import interactive map, roles, and contexts to construct the complete scene. We expect that learner can enhance the learning effectiveness to learning Taiwanese history and geography, as well as their learning motivation. The entertaining, challenging, rules, and interactive feedbacks of digital games can be transformed into learning motivations.

Keywords- digital learning game, role-play game, game design, Oblivion, Taiwan history and geography

I. INTRODUCTION

Prensky [31] said that game is the learning method the brain likes the most. Game has been known to take an important role in human activities, and is the most natural way for human to build knowledge and skills [7]. Because digital games can involve players in its amusement, it can intrigue learning motivation, enhance active participations by its nature characteristics. At the same time, it can help to enhance learners' self-directed learning and problem-solving abilities [2]. Entering the digital era, games have transformed itself from manipulative tools into virtual reality. However, the categorization of digital games is complicated and subjective that can be sorted out as many as dozens. It can be classified based on platforms, number of players, player types, subject matter, and sociality. Although most games emphasize on its entertaining feature, they more or less embed learning functions and implications. Game can contribute to the educational realm since it can intrigue learners' motivation and promote active participation to learners [18] [25]. Therefore, integrate digital games in classroom teaching can offer the opportunities for learners to develop both their critical and creative thinking abilities.

Current digital games have both entertaining and educational functions. It can successfully transfer traditional lecture-type education into learner-based learning model. Computer game software is also effective learning tools to remedial education and teaching about complicated subjects. Its multimedia presentation has strong attraction to users. It has been speculated that computer game play by young people has negative correlates or consequences, although little evidence has emerged to support these fears [11].

No evidence was obtained of negative outcomes among game players. On several measures—including family closeness, activity involvement, positive school engagement, positive mental health, substance use, self-concept, friendship network, and disobedience to parents—game players scored more favorably than did peers who never played computer games [11].

However, to create a complete instructional design to digital game and a set of useful evaluation criteria to students' learning results have to integrate the art of instructional design and science of technology. Analyses of the cognitive and perceptual-motor aspects of the activity indicate that, rather than being an intellectually lazy pursuit, computer game play requires high levels of skill and players elect to meet increasing challenges [14] [15] [39]. Klabbers [21] described that playing a computer game is an activity involving skill, knowledge or chance, in which players follow fixed rules to solve a puzzle or an organized event, in which competition involving several strategies will take place. In the pre-designed context of instructional games, the students explore, think and make decisions [26].

To integrate learning materials into games does not only require good script, production techniques, character design, environment modeling, but also knowledge management ability. Currently, there are many researches that try to integrate digital games in teachings of various subjects, such as math (e.g. [6]), biology (e.g. [12]), and computer science (e.g. [30]) and find that games can have better learning effects than traditional teaching, especially for subjects that are complicated. Therefore, this research tries to use digital

games to allow students to experience the histories and geographies that they cannot experience in person.

The overall picture that emerges from the present pattern of findings is that computer game play is one manifestation of an active and well-adjusted lifestyle. Rather than disrupting family life, games are played by young people who tend to perceive their family relations as close. Rather than displacing activities such as clubs and sports, games appear to be another leisure pursuit of those who are active social participants. Rather than signifying academic problems, game play is associated with more positive engagement with school [11].

The purpose of this research is to develop a digital game using 3D role-play style. The game context is a local history about the anti-Japanese war in Qing dynasty. The seven stages of game is based on the seven major locations in accordance to the war history. It is our goal to create a lively and interactive multimedia environment to allow students to learn history in the interesting and challenging way. At the same time to enhance their learning effectiveness and motivations. The game built in this research was done with Oblivion game engine. First, we design a role-play game script according to the anti-Japanese war history and geography of Southern Taiwan. And then we import interactive map, roles, and contexts to construct the complete scene. We expect that learner can enhance the learning effectiveness to learning Taiwanese history and geography, as well as their learning motivation. The entertaining, challenging, rules, and interactive feedbacks of digital games can be transformed into learning motivations.

II. LITERATURE REVIEW

The ideology of experiential learning provides a fruitful basis for integration of gameplay and pedagogy. In fact, Gredler [13] has categorized simulations and learning games for experiential training. For example, business games are based on experiential learning theory [19] [23] [28].

In order to build better connection between digital games and education, “serious games” were widely developed to enhance the games’ educational value; but unfortunately it sacrifices their entertaining value when the form of game becomes too “serious”. Hämäläinen [18] suggested that the design of digital games needs to consider both theoretical and practical needs as well as the potentiality of extending current games’ value.

Basically, most computer games can be transferred as problem solving instruments [32]. Games can be categorized into several types including car racing games, fighting games, sports games, action games, role-playing games, strategy games, simulated games, and so forth. Among which, only role-playing games, strategy games, and simulated games have more educational elements in them. Role-playing games (RPGs) are used most often both on the market and in education. RPGs emphasize on the growth and experience of the characters in the games. On the market, we often see games in middle-age European context and those characters have either challenges to conquer or tasks to achieve. Players can be one or more characters in the game, and manipulate the personal features of those characters. Throughout the

gaming process, the characters would grow up and gain experiences as they move, interact with others and the environment, equip, and gain or lose energies.

Generally, games provide a meaningful environment for problem-based learning. The ability to solve problems is one of the most important features of human skills [17]. Thus, one goal of education is to groom students to encounter novel situations [4]. Problem solving can be regarded as striving toward a goal which is not immediately attainable. Games provide a meaningful framework for offering problems to students. In fact, a game itself is a big problem that is composed of smaller causally linked problems. The nature of challenges that constitute the problem can vary greatly. Generally, a problem can be anything that somehow restricts a player’s progress in the game world [20]

Games have two major elements, characters that can be manipulated and a lot of stories. Supporting narrative in computer-based systems is essential [27] [9]. Players can experience what they do not have in real life and project their affections in it. They face problems, get tasks from the story, and acquire equipments through actions and fighting. Because the characters can greatly influence the effects of the whole game, game designers have to create characters which the players can identify with.

For example, simulation games offer possibilities to students to interact with the game by exploring and manipulating objects in order to test their hypotheses. Thus, while experiencing the game world, students become active participants in the learning processes and their motivation may shift from extrinsic to intrinsic rewards [5].

Other than that, role-playing games consist of map (maze), free actions (movements), and adventures. Therefore, the main design point is the development and exploration of the characters. Rollings and Adams [35] point out that the key to the role-playing game are storylines and character developments. Although stories are sometimes neglected by the designers, but the development, adventure, and fighting of the characters are necessary. Themes, backgrounds, interaction models such as character management, guidance, control, and objects, as well as viewpoints are essential to the design.

Game should be usable and provide clear goals and appropriate feedback to the player in order to facilitate flow experience. Additionally, according to Hoffman and Novak [16], focused attention is an important factor that contributes to the flow experience. The reflective observation of the feedback may lead to the construction of schemata and enable the discovery of new and better solutions to the problems. In online learning environments reflection has been facilitated for example with conversation tools, intelligent tutorials [37] and computer-based tutors [1] that can be utilized also in educational games.

Flow describes a state of complete absorption or engagement in an activity and refers to the optimal experience [10]. During optimal experience, a person is in a psychological state where he or she is so involved with the goal driven activity that nothing else seems to matter. Past research has shown that the flow state has positive impact on learning [40] and should be taken into account when

designing digital learning materials including games [20]. Online instruction, including games, should also support creativity and involve critical thinking [24]. According to Costkyan [8] good gameplay keeps a player motivated and engaged throughout an entire game. Moreover, instantiating real-time assessments & feedback as part of the training tasks can build metacognitive skills such as analyzing and assessing decision making processes [33] [34].

Generally, games satisfy the basic requirements of learning environments identified by Norman [29] and can provide an engaging environment for learning [20]. The use of stories in games is a fundamental part of game design [35]. Computer games may create a new learning culture that corresponds better with students' habits and interests [31]. Educators have tried to develop learning environments that support problem solving in complex life-like situations [38]. Games are designed to generate a positive affect in players and are most successful and engaging when they facilitate the flow experience [20].

A game is an exercise of voluntary control systems, in which there is a contest between powers, confined by rules in order to produce a disequilibrium outcome [3]. Salen and Zimmerman [36] defined game play as the formal interaction that ensues when players follow rules and structures that have been designed to result in an experience.

Experiential learning theory has long inspired the designers of digital learning environments [23]. Experiential learning builds upon the work of Piaget, Lewin, and Dewey [28]. Experiential learning theory consists of several models that stress the importance of direct experience and reflective observation. Kolb's [22] experiential learning model that consists of four stages is the central work in the field of experiential learning. According to the model, learning begins with a concrete experience followed by collection of data and reflective observations about that experience. In the abstract conceptualization stage a learner makes generalizations, draws conclusions, and forms hypotheses about the experience. In the final stage, the learner tests these hypotheses and ideas through active experimentation in new circumstances. Generally, the model stresses the continuous nature of learning and the appropriate feedback which provides the basis for a continuous process of goal-directed action.

Game production includes five stages:

a. Game theme and background setup: In this stage, the production team needs to create an attractive game story and context, and define game genre.

b. Game map setup: In this stage, the production team needs to design the game scenario and build up architectures related to it.

c. Game stages and tasks: In this stage, the production team needs to define content and gaming route with their corresponding challenges and tasks.

d. Game characters and object parameters: In this stage, the production team needs to define characters' appearances, personalities, and actions as well as tools and objects' functions and parameters.

e. Game mechanism and algorithm: In this stage, the production team needs to setup time system, fighting system,

support system, game interface and manipulation system, and related functions.

The production of the game in this research is based on these five stages and realizes the basic elements of game including fun, challenge, and learning ability.

III. RESEARCH METHOD

The development of game has been categorized into five stages.

a. Game context design. In this stage, game scripts are designed using the history of Southern Taiwan Anti-Japanese War, and role-play game is chosen to be the genre used.

b. Game map implementation. In this stage, architecture models are built using 3ds Max, and simulated environment is established with the game engine.

c. Game stages and tasks set up. The game stages were setup according to the locations of war history, and corresponding learning tasks were defined in each stage.

d. Game roles and objects set up. Game characters, objects, and properties parameters were defined according to the storyline.

e. Game mechanisms and algorithms set up. System operation mechanism and time as well as system functions and interface were set up to enable the activation of game.

The game production procedure is as Figure 1.

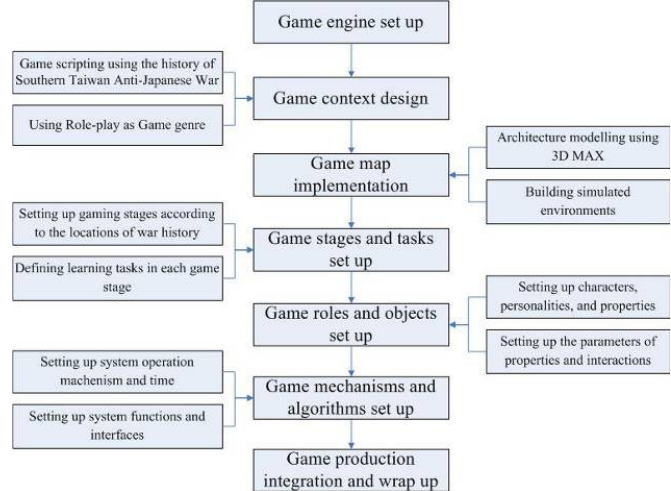


Figure 1. Game production procedure

IV. GAME DEVELOPMENT

This game used Wan Tan portion of Qian Long Taiwan Map as the location base which is most faithful to the time period of the story (Figure 2).

According to the story, there are about 20 character types in the game including Taiwanese, Hakka, aboriginal tribes, Japanese people and soldiers, as well as volunteer troops.

Seven main locations were chosen as the game stages. The main character of the game will progress following the story route. They are Wan Tan, Dong Gang, Wan Luan, Ping Tung, Chao Zhou, Da Wu Shan, and Hou Bi Lin.

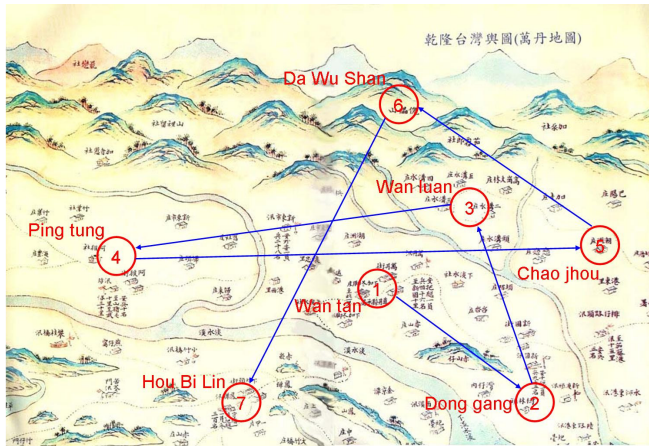


Figure 2. Game map and seven stages according to the locations of war history (Qian Long Taiwan Map)

The content and tasks of the game stages is in accordance to the social science learning objectives of the primary and secondary education. The main learning content includes:

- Understand the natural environment and atmosphere condition of Taiwan, and analyze the impact of the over-use of natural resources to the country.
- Understand the hardship of opening the wasteland and appreciate the Taiwanese culture.
- Understand the historic development of Taiwan from the Paleolithic Age, Iron Age, to Modern Age of the aboriginal tribe live.
- Understand the social, cultural, economic, and political development of Taiwan from The Great Voyage Age, Ming Dynasty, to Qing Dynasty.
- Understand the human environment of Taiwan.
- Understand the Japanese Colonial time of Taiwan.
- Understand the social lives of Taiwan.

The game stages are linked by narrations and animations.

Gamers can follow the plot to progress or jump between stages by using the main map. All tasks in the stages were also designed according to the social science capability index of schools.

This game used Oblivion as the game engine for the development. Oblivion has many pre-installed and player-shared objects, such as architectural and agricultural materials, character examples, properties, costumes, and action scripts, for designers to use. Using this game engine, designers can effectively and successfully build the scenarios and characters in Taiwanese styles. At the same time, martial art stories and actions can be easily established so that players can be immersed in the historical stories and the war fighting atmosphere.

There are four major features and functions of Oblivion:

- Character and costume design. Figure 3 shows one of the main characters "Lin Cat Dad".
- Interior design and property setup. Figure 4 shows Wan-Jin Catholic Cathedral which is one of the main buildings in the game.

c. Scenario and environment setup. Figure 5 shows the Sakura Shop in Dong Gang which is one of the main buildings created in the story.

d. High degree of action manipulation and simulated physics logic and combat training system. Figure 6 shows a combat scene which simulates the martial art fights for the game.



Figure 3. Character and Costume Design



Figure 4. Interior Design and Property Set Up



Figure 5. Scenario and Environment Set Up



Figure 6. Action and Combat System

We use 3ds Max to build the models of buildings and character costumes. All major buildings were built as real as the true architecture (Figure 7). By doing this, students can experience three dimensional scenarios and environments of the historic time which cannot be done with textbook teaching.



Figure 7. Real Architecture (left) and 3D Architectural Model Built with 3ds Max (right)

The production of the game is done with Oblivion. The game world is created with the real world in terms of the area

ratio, land shape, and heights. The plants were planted according to the heights, areas, and seasons. The roads and major land spots of the towns were planned and modeled as authentic as possible.

In this game, all characters, creatures, and properties have their own parameters. Characters are imposed with certain personalities that they live in the simulated environment as in the real world. Players interact and have dialogues with the non-player-character (NPC), and get feedbacks. All personalities of the characters can trigger different situations when encounter different characters and objects in different times (Figure 8). Therefore, no identical situation would happen between every entrance to the world.



Figure 8. Interaction with NPC

During the loading of every game program, small pieces of learning content are shown to allow more learning inputs. For example, the meaning of Da Wu Mountain, originality of Zhenhai Temple in Dong Kang, and the origin of Sung-Jiang Martial Art Formation, etc.

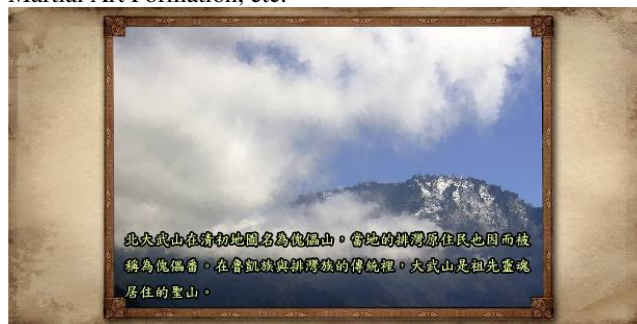


Figure 9. Learning content shown while loading

V. CONCLUSION

The purpose of this research is to develop a digital game using 3D role-play style. The game context is a local history about the anti-Japanese war in Qing dynasty. It is hoped to let the players to experience the time and place of the history, and learn with fun.

In the future, we will first evaluate the usability of the game, and then conduct educational experiment. We will conduct pre- and post-tests which was checked and modified by social science teachers. They are to evaluate students' learning achievements after playing the game. At the same time, questionnaire with learning motivation, learning attitudes and satisfactions will be performed along with focus group interviews. Both quantitative and qualitative data will be collected to analyze the learning effectiveness of the game.

We expect that digital learning games can effectively increase learning effectiveness since the game content is not imaginary. The game presented a simulated world to the real histories and geographies of Taiwan. We anticipate that the entertaining effects, challenges, interactions, and feedbacks of the game can enhance the learning motivation to history and geography of students.

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REFERENCES

- [1] Aleven, V., & Koedinger, K. R. (2002). An effective metacognitive strategy: learning by doing and explaining with a computer-based cognitive tutor. *Cognitive Science*, 26, 147–179.
- [2] Amory, A., Naicker, K., Vincent, J., & Adams, C. (1999). The use of computer games as an educational tool: Identification of appropriate game types and game elements. *British Journal of Educational Technology*, 30(4), 311–322.
- [3] Avedon, E. M., & Sutton-Smith, B. (1971). *The study of games*. New York: John Wiley and Sons.
- [4] Bruer, J. T. (1993). *Schools for thought: A science of learning in the classroom*. Cambridge MIT Press.
- [5] Bruner, J. S. (1961). The act of discovery. *Harvard Educational Review*, 31, 21–32.
- [6] Cai, Y. Y., Indhumathi, C., Chen, W. Y., & Zheng, J. M. (2008). VR Bio X Games. *Lecture Notes in Computer Science*, 5080, 278–287.
- [7] Cai, Y. -Y., Lu, B. -F., Fan, Z. -W., Indhumathi, C., Lim, K. -T., Chan, C. -W., Jiang, Y., & Li, L. (2006). Bio-edutainment: learning life science through X gaming. *Computers & Graphics*, 30, 3–9.
- [8] Costkian, G. (2002). I have no words & I must design: toward a critical vocabulary for games. *Proceedings of the computer games and digital cultures conference, Finland*.
- [9] Crawford, C. (2005). *Chris Crawford on Interactive Storytelling*. New Riders Publishing, Indianapolis, IN.
- [10] Csikszentmihalyi, M. (1991). *Flow: The psychology of optimal experience*. New York Harper Perennial.
- [11] Durkin, K., & Barber, B. (2002). Not so doomed: computer game play and positive adolescent development. *Journal of Applied Developmental Psychology*, 23, 373–392.
- [12] Fengfeng, K., & Barbara, G. (2007). Gameplaying for maths learning: cooperative or not? *British Journal of Educational Technology*, 38(2), 249–259.
- [13] Gredler, M. E. (1996). Educational games and simulations: a technology in search of a (research) paradigm. In D. H. Jonassen (Ed.), *Handbook of research for educational communications and technology*. New York Simon & Schuster Macmillan.
- [14] Greenfield, P. M. (1984). *Mind and media: The effects of television, video games and computers*. London: Cambridge, MA: Harvard University Press.
- [15] Greenfield, P. M. (1994). Video games as cultural artifacts. *Journal of Applied Developmental Psychology*, 15, 3–12.
- [16] Hoffman, D. L., & Novak, T. P. (1996). Marketing in hypermedia computer-mediated environments: Conceptual foundations. *Journal of Marketing*, 60, 50–68.
- [17] Holyoak, K. J. (1991). Problem solving. In D. N. Osherson, & E. E. Smith (Eds.), *Thinking. An invitation to cognitive science* (pp. 117–146). Cambridge The MIT Press.
- [18] Hämäläinen, R. (2008). Designing and evaluating collaboration in a virtual game environment for vocational learning. *Computers & Education*, 50(1), 98–109.
- [19] Isaacs, W., & Senge, P. (1992). Overcoming limits to learning in computer-based learning environments. *European Journal of Operational Research*, 59, 183–196.
- [20] Kiili, K. (2005). Digital game-based learning: Towards an experiential gaming model. *The Internet and Higher Education*, 8, 13–24.
- [21] Klabbers, J. H. G. (1999). *Three easy pieces: a taxonomy on gaming*. London: Kogan Page.
- [22] Kolb, D. (1984). *Experiential learning: Experience as the source of learning and development*. New Jersey Prentice Hall.
- [23] Lainema, T. (2003). *Enhancing organizational business process perception: Experiences from constructing and applying a dynamic business simulation game*. PhD thesis, Turku School of Economics and Business Administration.
- [24] MacDonald, C. J., Stodel, E. J., Farres, L. G., Breithaupt, K., & Gabriel, M. A. (2001). The demand-drive learning model: a framework for web-based learning. *Internet and Higher Education*, 4, 9–30.
- [25] Mann, B. D., Eidelson, B. M., Fukuchi, S.G., Nissman, S.A., Robertson, S., & Jardines, L. (2002). The development of an interactive game-based tool for learning surgical management algorithms via computer. *The American Journal of Surgery*, 183(3), 305–308.
- [26] Mayer, R. E. (1991). *Thinking, problem solving, cognition (2nd ed)*. NY: Freeman.
- [27] Murray, J., 1997. *Hamlet on the Holodeck: The Future of Narrative in Cyberspace*. MIT Press, Cambridge, MA.
- [28] Nielsen-Englyst, L. (2003). Game design for imaginative conceptualisation. *Proceedings of the international workshop on experimental interactive learning in industrial management, Allborg* (pp. 149–164).
- [29] Norman, D. A. (1993). *Things that make us smart: Defending human attributes in the age of the machine*. New York Addison-Wesley.
- [30] Papastergiou, M. (2009). Digital Game-Based Learning in high school Computer Science education: Impact on educational effectiveness and student motivation. *Computers & Education*, 52, 1–12.
- [31] Prensky, M. (2001). *Digital game-based Learning*. New York: McGraw-Hill.
- [32] Rao, S. S. (1998). Putting fun back into learning. *Training*, 44–55.
- [33] Raybourn, E.M. (2005a). *Adaptive thinking and leadership: New ways to assess in-game actions*. Presented at Serious Games Summit, Washington, D.C.
- [34] Raybourn, E.M. (2006). Simulation experience design methods for training the forces to think adaptively. I/ITSEC 2006 Proceedings, Interservice/Industry Training, Simulation and Education Conference, USA.

- [35] Rollings, A., & Adams, E. (2003). *Andrew Rollings and Ernest Adams on game design*. USA New Riders.
- [36] Salen, K., Zimmerman, E. (2004). *Rules of Play*. The MIT Press, Cambridge, MA.
- [37] Seale, J. K., & Cann, A. J. (2000). Reflection on-line or off-line: the role of learning technologies in encouraging students to reflect. *Computers & Education*, 34, 309–320.
- [38] Suomala, J. (1999). *Students' problem solving in the Lego/Logo learning environment*. PhD thesis, University of Jyvaskyla.
- [39] Turkle, S. (1984). *The second self*. Computers and the human spirit. London: Granada.
- [40] Webster, J., Trevino, L. K., & Ryan, L. (1993). The dimensionality and correlates of flow in human-computer interaction. *Computers in Human Behavior*, 9, 411–426.